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Ai Unit 5 - AI notes for unit 5

Artificial Intelligence (Dr. Babasaheb Ambedkar Technological University)

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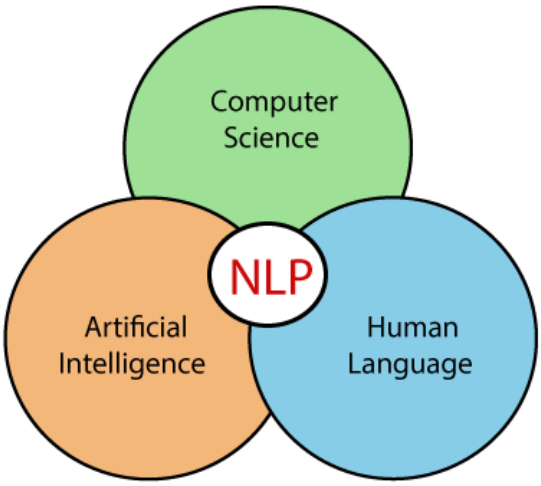
Unit 5: Natural Language Processing

What is NLP?

NLP stands for Natural Language Processing, which is a part of Computer Science, Human language, and Artificial Intelligence. It is the technology that is used by machines to understand, analyse, manipulate, and interpret human's languages. It helps developers to organize knowledge for performing tasks such as translation, automatic summarization, Named Entity Recognition (NER), speech recognition, relationship extraction, and topic segmentation.



Processing of Natural Language is required when you want an intelligent system like robot to perform as per your instructions, when you want to hear decision from a dialogue based clinical expert system, etc.

The field of NLP involves making computers to perform useful tasks with the natural languages humans use. The input and output of an NLP system can be

- **Speech**
- **Written Text**

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Components of NLP

There are the following two components of NLP -

1. Natural Language Understanding (NLU)

Natural Language Understanding (NLU) helps the machine to understand and analyse human language by extracting the metadata from content such as concepts, entities, keywords, emotion, relations, and semantic roles.

NLU mainly used in Business applications to understand the customer's problem in both spoken and written language.

NLU involves the following tasks -

- It is used to map the given input into useful representation.
- It is used to analyze different aspects of the language.

The next and hardest step of NLP is the understanding part.

- First, the computer must comprehend the meaning of each word. It tries to figure out whether the word is a noun or a verb, whether it's in the past or present tense, and so on. This is called Part-of-Speech tagging (POS).
- A lexicon (a vocabulary) and a set of grammatical rules are also built into NLP systems. The most difficult part of NLP is understanding.
- The machine should be able to grasp what you said by the conclusion of the process. There are several challenges in accomplishing this when considering problems such as words having several meanings (polysemy) or different words having similar meanings (synonymy), but developers encode rules into their NLU systems and train them to learn to apply the rules correctly.

2. Natural Language Generation (NLG)

Natural Language Generation (NLG) acts as a translator that converts the computerized data into natural language representation. It mainly involves Text planning, Sentence planning, and Text Realization.

NLG is much simpler to accomplish. NLG converts a computer's machine-readable language into text and can also convert that text into audible speech using text-to-speech technology.

- First, the NLP system identifies what data should be converted to text. If you asked the computer a question about the weather, it most likely did an online search to find your answer, and from there it decides that the temperature, wind, and humidity are the factors that should be read aloud to you.
- Then, it organizes the structure of how it's going to say it. This is similar to NLU except backward. NLG system can construct full sentences using a lexicon and a set of grammar rules.
- Finally, text-to-speech takes over. The text-to-speech engine uses a prosody model to evaluate the text and identify breaks, duration, and pitch. The engine then combines all the recorded phonemes into one cohesive string of speech using a speech database.

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Difference between NLU and NLG

NLU	NLG
NLU is the process of reading and interpreting language.	NLG is the process of writing or generating language.
It produces non-linguistic outputs from natural language inputs.	It produces constructing natural language outputs from non-linguistic inputs.

Advantages of NLP

- NLP helps users to ask questions about any subject and get a direct response within seconds.
- NLP offers exact answers to the question means it does not offer unnecessary and unwanted information.
- NLP helps computers to communicate with humans in their languages.
- It is very time efficient.
- Most of the companies use NLP to improve the efficiency of documentation processes, accuracy of documentation, and identify the information from large databases.

Disadvantages of NLP

A list of disadvantages of NLP is given below:

- NLP may not show context.
- NLP is unpredictable
- NLP may require more keystrokes.
- NLP is unable to adapt to the new domain, and it has a limited function that's why NLP is built for a single and specific task only.

Some common roles in Natural Language Processing (NLP) include:

- NLP engineer: designing and implementing NLP systems and models
- NLP researcher: conducting research on NLP techniques and algorithms
- ML engineer: Designing and deployment of various machine learning models including NLP.
- NLP data scientist: analyzing and interpreting NLP data
- NLP consultant: providing expertise in NLP to organizations and businesses.

Phases of NLP

There are the following five phases of NLP:

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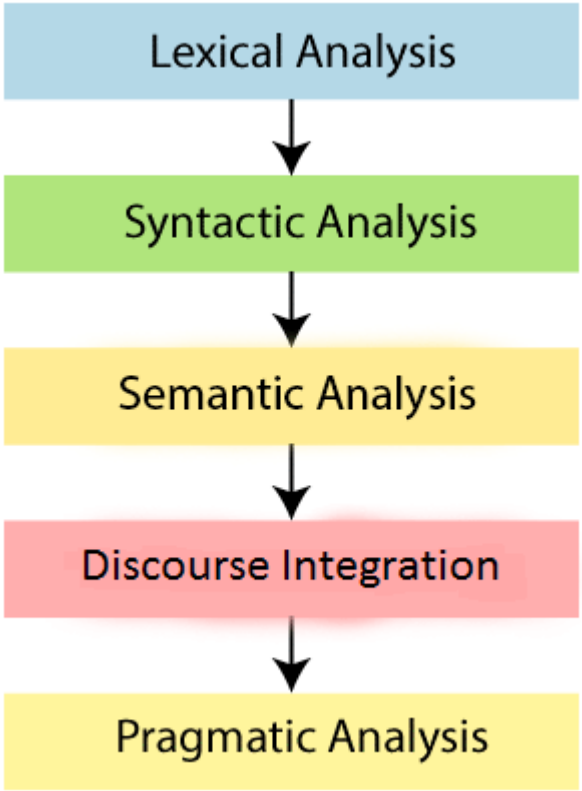

1. Lexical Analysis and Morphological

The first phase of NLP is the Lexical Analysis. This phase scans the source code as a stream of characters and converts it into meaningful lexemes. It divides the whole text into paragraphs, sentences, and words.

It involves identifying and analyzing the structure of words. Lexicon of a language means the collection of words and phrases in a language. Lexical analysis is dividing the whole chunk of txt into paragraphs, sentences, and words.

2. Syntactic Analysis (Parsing)

Syntactic Analysis is used to check grammar, word arrangements, and shows the relationship among the words. It involves analysis of words in the sentence for grammar and arranging words in a manner that shows the relationship among the words. The sentence such as “The school goes to boy” is rejected by English syntactic analyzer.

Example: Agra goes to the Poonam

In the real world, Agra goes to the Poonam, does not make any sense, so this sentence is rejected by the Syntactic analyzer.

3. Semantic Analysis

Semantic analysis is concerned with the meaning representation. It mainly focuses on the literal meaning of words, phrases, and sentences. It draws the exact meaning or the dictionary meaning from the text. The text is checked for meaningfulness. It is done by mapping syntactic structures and objects in the task domain. The semantic analyzer disregards sentence such as “hot ice-cream”.

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4. Discourse Integration

Discourse Integration depends upon the sentences that proceeds it and also invokes the meaning of the sentences that follow it. he meaning of any sentence depends upon the meaning of the sentence just before it. In addition, it also brings about the meaning of immediately succeeding sentence.

5. Pragmatic Analysis

Pragmatic is the fifth and last phase of NLP. It helps you to discover the intended effect by applying a set of rules that characterize cooperative dialogues. During this, what was said is re-interpreted on what it actually meant. It involves deriving those aspects of language which require real world knowledge.

For Example: "Open the door" is interpreted as a request instead of an order.

Applications of NLP

There are the following applications of NLP -

1. Question Answering

Question Answering focuses on building systems that automatically answer the questions asked by humans in a natural language.

2. Spam Detection

Spam detection is used to detect unwanted e-mails getting to a user's inbox.

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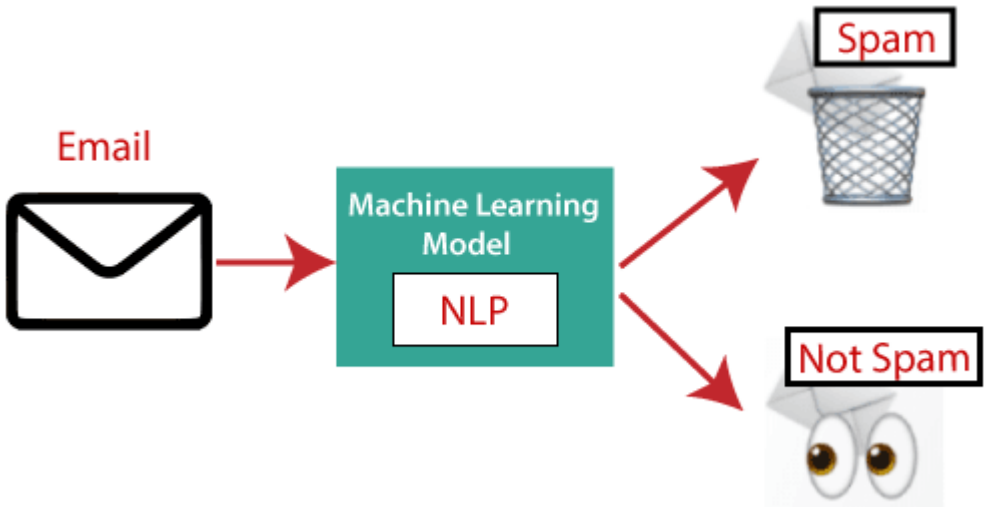
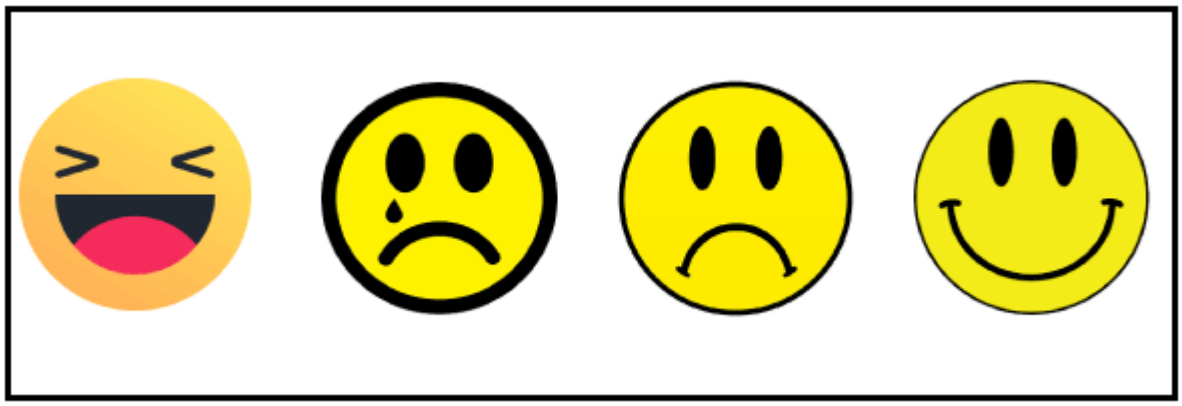


3. Sentiment Analysis

Sentiment Analysis is also known as **opinion mining**. It is used on the web to analyse the attitude, behaviour, and emotional state of the sender. This application is implemented through a combination of NLP (Natural Language Processing) and statistics by assigning the values to the text (positive, negative, or natural), identify the mood of the context (happy, sad, angry, etc.)



4. Machine Translation

Machine translation is used to translate text or speech from one natural language to another natural language.

Example: Google Translator

5. Spelling correction

Microsoft Corporation provides word processor software like MS-word, PowerPoint for the spelling correction.

6. Speech Recognition

Speech recognition is used for converting spoken words into text. It is used in applications, such as mobile, home automation, video recovery, dictating to Microsoft Word, voice biometrics, voice user interface, and so on.

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7. Chatbot

Implementing the Chatbot is one of the important applications of NLP. It is used by many companies to provide the customer's chat services.

8. Information extraction

Information extraction is one of the most important applications of NLP. It is used for extracting structured information from unstructured or semi-structured machine-readable documents.

9. Natural Language Understanding (NLU)

It converts a large set of text into more formal representations such as first-order logic structures that are easier for the computer programs to manipulate notations of the natural language processing.

Form of Learning

1. **Supervised Learning:**
    - In supervised learning, the algorithm is trained on a labeled dataset, which means that it learns from input-output pairs. It tries to map inputs to correct outputs.
    - Common applications include classification and regression tasks, such as image recognition and predicting house prices.
2. **Unsupervised Learning:**
    - Unsupervised learning involves training an algorithm on unlabeled data, and the goal is to find patterns or structures within the data.
    - Clustering and dimensionality reduction are typical unsupervised learning tasks, as seen in customer segmentation and feature extraction.
3. **Semi-Supervised Learning:**
    - Semi-supervised learning combines elements of both supervised and unsupervised learning. It uses a small amount of labeled data and a large amount of unlabeled data for training.
    - It's useful when labeling data is expensive or time-consuming.
4. **Reinforcement Learning:**
    - Reinforcement learning involves an agent that learns to interact with an environment to maximize a reward. It learns by trial and error, receiving feedback in the form of rewards or penalties.
    - Applications include game playing, robotics, and autonomous systems.
5. **Self-Supervised Learning:**
    - Self-supervised learning is a type of unsupervised learning where the algorithm creates its own labels or tasks from the data.
    - It's often used for pre-training deep neural networks, where the model learns to predict parts of the input data from other parts.
6. **Transfer Learning:**
    - Transfer learning leverages knowledge gained from one task to improve performance on another related task.

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- Pre-trained models, such as BERT for natural language processing, are widely used examples.

7. **Meta-Learning:**
    - Meta-learning focuses on teaching models to learn how to learn. It involves training models to adapt quickly to new tasks.
    - This can be useful in situations where an AI system needs to continually learn and adapt.
8. **Online Learning:**
    - In online learning, models are updated continuously as new data becomes available. It is well-suited for applications where data streams in real-time.
    - Online learning can be used in recommendation systems and fraud detection.
9. **Few-Shot and Zero-Shot Learning:**
    - Few-shot and zero-shot learning involve training models to make predictions with very few or no examples from a particular class.
    - This is valuable when data is scarce, as seen in identifying rare objects or languages.
10. **Heuristic Search:**

- Heuristic search methods use domain-specific knowledge or rules to guide the learning process.
- These methods are common in game playing AI, such as chess and Go.

Inductive Learning

- Inductive Learning Algorithm (ILA) is an iterative and inductive machine learning algorithm that is used for generating a set of classification rules, which produces rules of the form "IF-THEN", for a set of examples, producing rules at each iteration and appending to the set of rules.
- There are basically two methods for knowledge extraction firstly from domain experts and then with machine learning. For a very large amount of data, the domain experts are not very useful and reliable. So we move towards the machine learning approach for this work. To use machine learning One method is to replicate the expert's logic in the form of algorithms but this work is very tedious, time taking, and expensive. So we move towards the inductive algorithms which generate the strategy for performing a task and need not instruct separately at each step.
- An technique of machine learning called inductive learning trains a model to generate predictions based on examples or observations. During inductive learning, the model picks up knowledge from particular examples or instances and generalizes it such that it can predict outcomes for brand-new data.
- When using inductive learning, a rule or method is not explicitly programmed into the model. Instead, the model is trained to spot trends and connections in the input data and then utilize this knowledge to predict outcomes from fresh data. Making a model that can precisely anticipate the result of subsequent instances is the aim of inductive learning.
- In supervised learning situations, where the model is trained using labeled data, inductive learning is frequently utilized. A series of samples with the proper output labels are used to train the model. The model then creates a mapping between the input data and the output data using this training data. The output for fresh instances may be predicted using the model after it has been trained.
- Inductive learning is used by a number of well-known machine learning algorithms, such as decision trees, k-nearest neighbors, and neural networks. Because it enables the development of models that can accurately anticipate new data, even when the underlying patterns and relationships are complicated and poorly understood, inductive learning is an essential method for machine learning.

Advantages

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- Because inductive learning models are flexible and adaptive, they are well suited for handling difficult, complex, and dynamic information.
- Finding hidden patterns and relationships in data: Inductive learning models are ideally suited for tasks like pattern recognition and classification because they can identify links and patterns in data that may not be immediately apparent to humans.
- Huge datasets − Inductive learning models are suitable for applications requiring the processing of massive quantities of data because they can efficiently handle enormous volumes of data.
- Appropriate for situations where the rules are ambiguous − Since inductive learning models may learn from examples without explicit programming, they are suitable for situations when the rules are not precisely described or understood beforehand.

Disadvantages

- May overfit to particular data − Inductive learning models that have overfit to specific training data, or that have learned the noise in the data rather than the underlying patterns, may perform badly on fresh data.
- computationally costly possible − The employment of inductive learning models in real-time applications may be constrained by their computationally costly nature, especially for complex datasets.
- Limited interpretability − Inductive learning models may be difficult to understand, making it difficult to understand how they arrive at their predictions, in applications where the decision-making process must be transparent and explicable.
- Inductive learning models are only as good as the data they are trained on, therefore if the data is inaccurate or inadequate, the model may not perform effectively.

Decision Tree Classification Algorithm

- Decision Tree is a **Supervised learning technique** that can be used for both classification and Regression problems, but mostly it is preferred for solving Classification problems. It is a tree-structured classifier, where **internal nodes represent the features of a dataset, branches represent the decision rules** and **each leaf node represents the outcome.**
- In a Decision tree, there are two nodes, which are the **Decision Node** and **Leaf Node.** Decision nodes are used to make any decision and have multiple branches, whereas Leaf nodes are the output of those decisions and do not contain any further branches.
- The decisions or the test are performed on the basis of features of the given dataset.
- ***It is a graphical representation for getting all the possible solutions to a problem/decision based on given conditions.***
- It is called a decision tree because, similar to a tree, it starts with the root node, which expands on further branches and constructs a tree-like structure.
- In order to build a tree, we use the **CART algorithm,** which stands for **Classification and Regression Tree algorithm.**
- A decision tree simply asks a question, and based on the answer (Yes/No), it further split the tree into subtrees.
- Below diagram explains the general structure of a decision tree:

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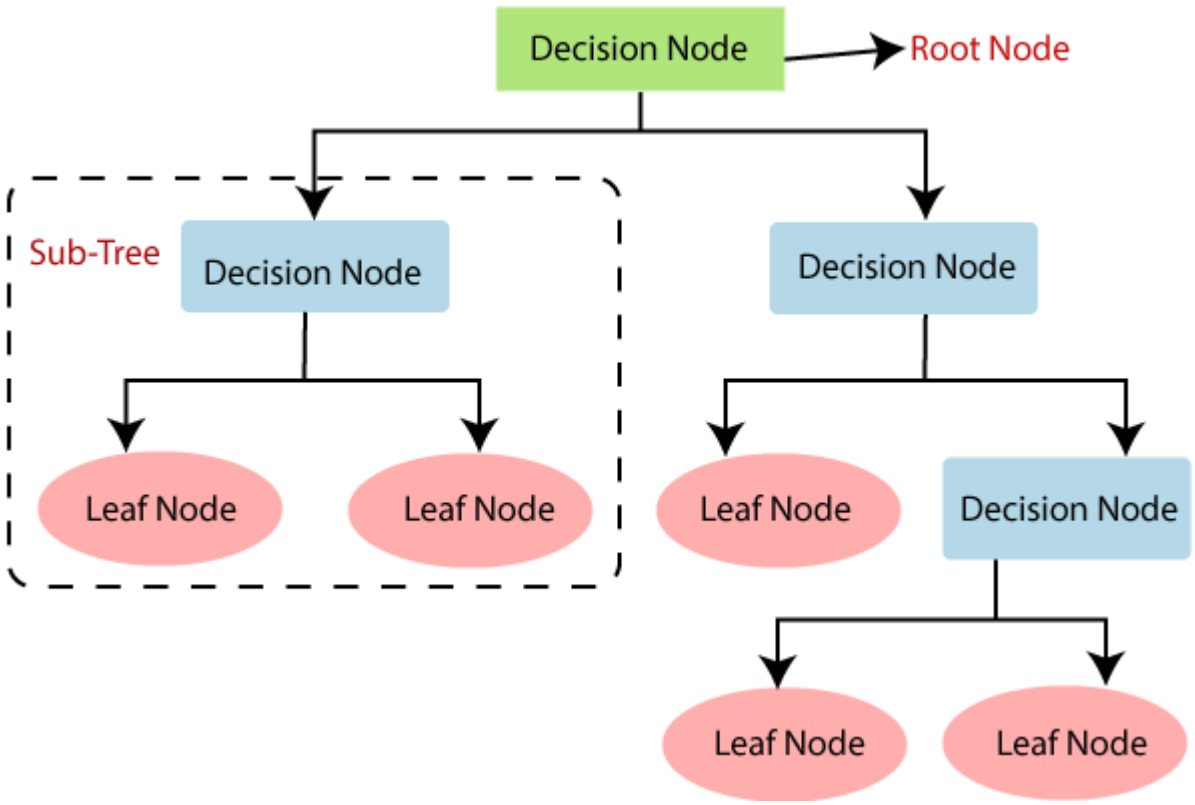

Why use Decision Trees?

There are various algorithms in Machine learning, so choosing the best algorithm for the given dataset and problem is the main point to remember while creating a machine learning model. Below are the two reasons for using the Decision tree:

- Decision Trees usually mimic human thinking ability while making a decision, so it is easy to understand.
- The logic behind the decision tree can be easily understood because it shows a tree-like structure.

Decision Tree Terminologies

- **Root Node:** Root node is from where the decision tree starts. It represents the entire dataset, which further gets divided into two or more homogeneous sets.
- **Leaf Node:** Leaf nodes are the final output node, and the tree cannot be segregated further after getting a leaf node.
- **Splitting:** Splitting is the process of dividing the decision node/root node into sub-nodes according to the given conditions.
- **Branch/Sub Tree:** A tree formed by splitting the tree.
- **Pruning:** Pruning is the process of removing the unwanted branches from the tree.
- **Parent/Child node:** The root node of the tree is called the parent node, and other nodes are called the child nodes.

How does the Decision Tree algorithm Work?

In a decision tree, for predicting the class of the given dataset, the algorithm starts from the root node of the tree. This algorithm compares the values of root attribute with the record (real dataset) attribute and,

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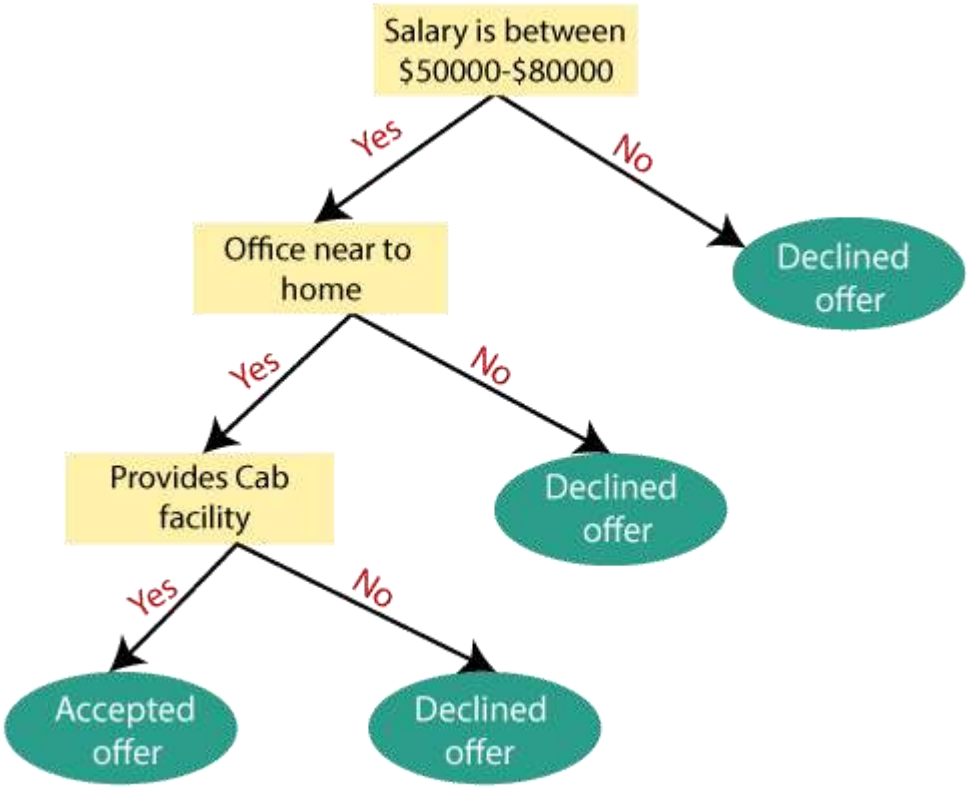
based on the comparison, follows the branch and jumps to the next node.

For the next node, the algorithm again compares the attribute value with the other sub-nodes and move further. It continues the process until it reaches the leaf node of the tree. The complete process can be better understood using the below algorithm:

- **Step-1:** Begin the tree with the root node, says S, which contains the complete dataset.
- **Step-2:** Find the best attribute in the dataset using **Attribute Selection Measure (ASM).**
- **Step-3:** Divide the S into subsets that contains possible values for the best attributes.
- **Step-4:** Generate the decision tree node, which contains the best attribute.
- **Step-5:** Recursively make new decision trees using the subsets of the dataset created in step -3. Continue this process until a stage is reached where you cannot further classify the nodes and called the final node as a leaf node.

Example: Suppose there is a candidate who has a job offer and wants to decide whether he should accept the offer or Not. So, to solve this problem, the decision tree starts with the root node (Salary attribute by ASM). The root node splits further into the next decision node (distance from the office) and one leaf node based on the corresponding labels. The next decision node further gets split into one decision node (Cab facility) and one leaf node. Finally, the decision node splits into two leaf nodes (Accepted offers and Declined offer). Consider the below diagram:



Explanation-based learning (EBL)

- Explanation-based learning is a type of machine learning that uses a very strong, or even flawless, domain theory to generalise or create ideas from training data. It is also tied to Encoding, which aids in learning.
- Explanation-Based Learning (EBL) is a moral strategy for improving supervised learning by utilising available domain knowledge. As a result, the rate of learning, the certainty of learning, the precision of the taught concept, or a combination of these can be enhanced.

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- It is a method for an intelligent system to learn by seeing instances. The capacity of EBl systems to generate justified generalisations from single training instances distinguishes them.

- Explanation-based learning can learn from just one training event. Rather than taking multiple examples, explanation-based learning emphasises learning a single, specific model. Take, for example, the Ludoo game. Explanation-based learning can learn from just one training event.

Explanation-based learning (EBL) is a form of learning in artificial intelligence (AI) that involves understanding the underlying domain or problem space and generating explanations for specific instances. Unlike other machine learning approaches that rely on numerous examples, EBL focuses on a deep understanding of individual examples, generating explanations based on underlying principles, and generalizing those explanations to cover other related instances.

Here's a more detailed overview of EBL:

1. **Analyzing the Problem:** EBL starts by analyzing a particular instance of a problem, applying domain knowledge to understand why a specific solution works.
2. **Generating an Explanation:** Based on the analysis, EBL generates an explanation of why the solution is correct. This explanation leverages domain knowledge, logical reasoning, and underlying principles to articulate why the solution makes sense.
3. **Generalizing the Explanation:** The explanation is then generalized to create a rule or a pattern that can be applied to other similar instances. This generalization allows the system to learn from a single example and apply the learning to many related cases.
4. **Storing the Generalized Rule:** The generalized rule is stored and used to solve similar problems in the future. By having this rule, the system can quickly solve related problems without having to analyze each one individually.
5. **Integration with Other Learning Techniques:** EBL can be integrated with other learning techniques, such as inductive learning and analogical reasoning, to form more comprehensive and robust learning models.

The advantage of EBL is that it can provide a deep understanding of specific problems, allowing for intelligent generalization to related instances. However, it requires significant domain knowledge and logical reasoning capabilities, making it a more complex approach than many other machine learning techniques.

Neural Network

The term "**Artificial Neural Network**" is derived from Biological neural networks that develop the structure of a human brain. Similar to the human brain that has neurons interconnected to one another, artificial neural networks also have neurons that are interconnected to one another in various layers of the networks. These neurons are known as nodes.

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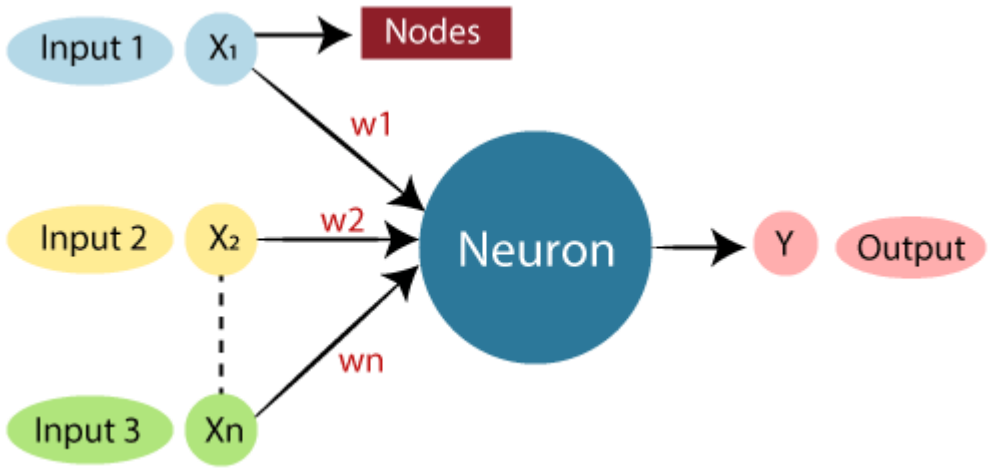
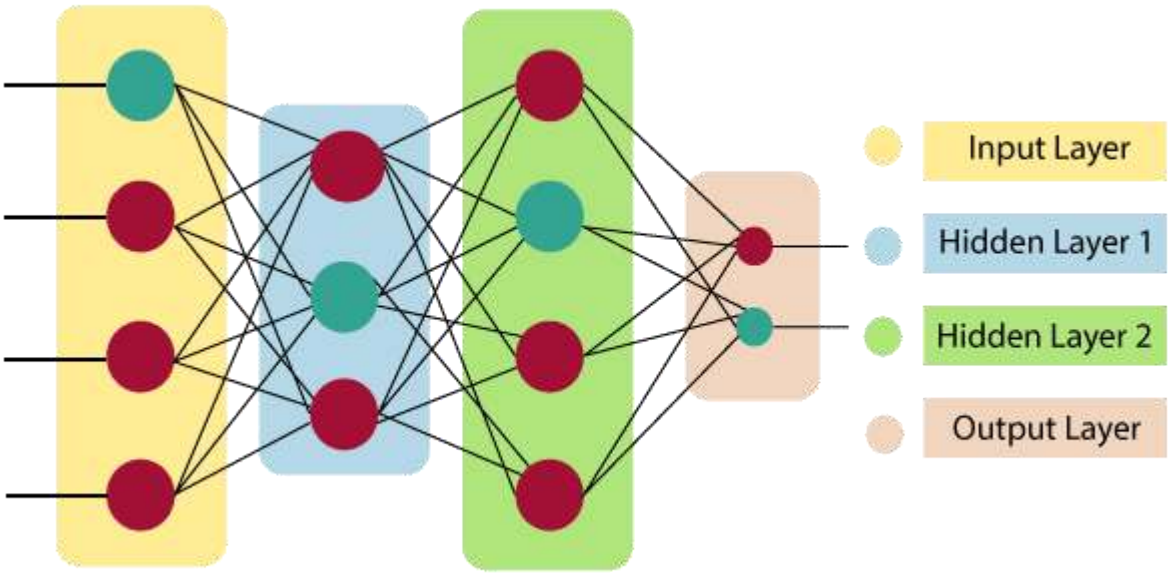


The architecture of an artificial neural network:

To understand the concept of the architecture of an artificial neural network, we have to understand what a neural network consists of. In order to define a neural network that consists of a large number of artificial neurons, which are termed units arranged in a sequence of layers. Lets us look at various types of layers available in an artificial neural network.

Artificial Neural Network primarily consists of three layers:



Input Layer:

As the name suggests, it accepts inputs in several different formats provided by the programmer.

Hidden Layer:

The hidden layer presents in-between input and output layers. It performs all the calculations to find hidden features and patterns.

Output Layer:

The input goes through a series of transformations using the hidden layer, which finally results in output

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that is conveyed using this layer.

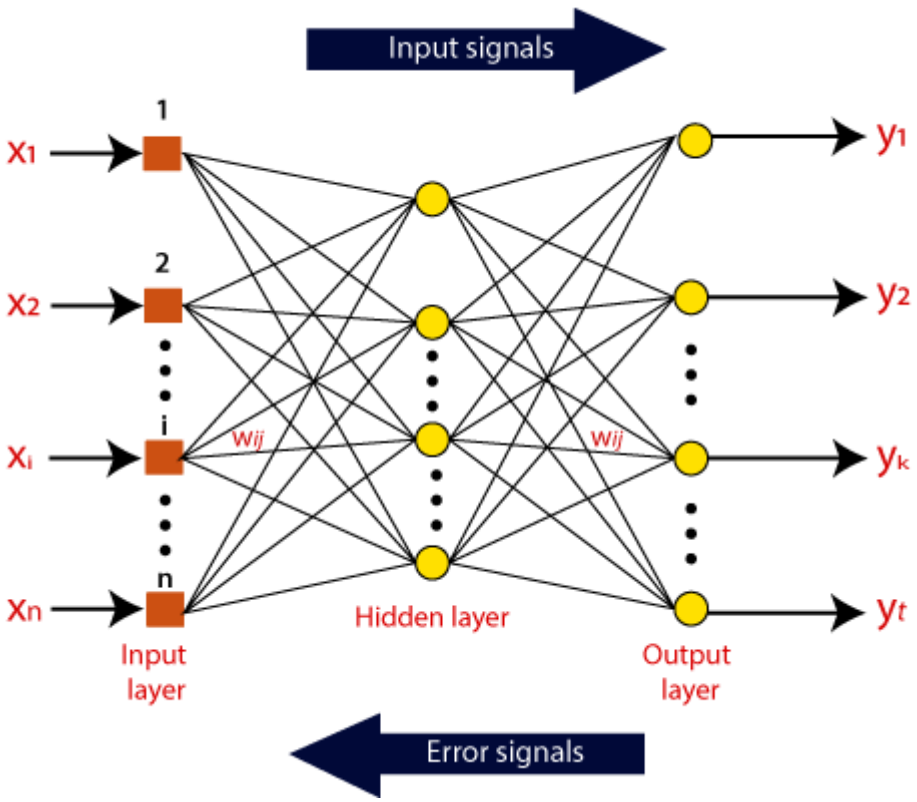
The artificial neural network takes input and computes the weighted sum of the inputs and includes a bias. This computation is represented in the form of a transfer function.

$$\sum_{i=1}^{n} Wi * Xi + b$$

It determines weighted total is passed as an input to an activation function to produce the output. Activation functions choose whether a node should fire or not. Only those who are fired make it to the output layer. There are distinctive activation functions available that can be applied upon the sort of task we are performing.

How do artificial neural networks work?

Artificial Neural Network can be best represented as a weighted directed graph, where the artificial neurons form the nodes. The association between the neurons outputs and neuron inputs can be viewed as the directed edges with weights. The Artificial Neural Network receives the input signal from the external source in the form of a pattern and image in the form of a vector. These inputs are then mathematically assigned by the notations x(n) for every n number of inputs.



Afterward, each of the input is multiplied by its corresponding weights (these weights are the details utilized by the artificial neural networks to solve a specific problem). In general terms, these weights normally represent the strength of the interconnection between neurons inside the artificial neural network. All the weighted inputs are summarized inside the computing unit.

If the weighted sum is equal to zero, then bias is added to make the output non-zero or something else to scale up to the system's response. Bias has the same input, and weight equals to 1. Here the total of

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weighted inputs can be in the range of 0 to positive infinity. Here, to keep the response in the limits of the desired value, a certain maximum value is benchmarked, and the total of weighted inputs is passed through the activation function.

The activation function refers to the set of transfer functions used to achieve the desired output. There is a different kind of the activation function, but primarily either linear or non-linear sets of functions. Some of the commonly used sets of activation functions are the Binary, linear, and Tan hyperbolic sigmoidal activation functions. Let us take a look at each of them in details:

Types of Artificial Neural Network:

There are various types of Artificial Neural Networks (ANN) depending upon the human brain neuron and network functions, an artificial neural network similarly performs tasks. The majority of the artificial neural networks will have some similarities with a more complex biological partner and are very effective at their expected tasks. For example, segmentation or classification.

Feedback ANN:

In this type of ANN, the output returns into the network to accomplish the best-evolved results internally. As per the University of Massachusetts, Lowell Centre for Atmospheric Research. The feedback networks feed information back into itself and are well suited to solve optimization issues. The Internal system error corrections utilize feedback ANNs.

Feed-Forward ANN:

A feed-forward network is a basic neural network comprising of an input layer, an output layer, and at least one layer of a neuron. Through assessment of its output by reviewing its input, the intensity of the network can be noticed based on group behavior of the associated neurons, and the output is decided. The primary advantage of this network is that it figures out how to evaluate and recognize input patterns.

Genetic learning

Genetic Algorithms(GAs) are adaptive heuristic search algorithms that belong to the larger part of evolutionary algorithms. Genetic algorithms are based on the ideas of natural selection and genetics. These are intelligent exploitation of random search provided with historical data to direct the search into the region of better performance in solution space. They are commonly used to generate high-quality solutions for optimization problems and search problems.

Genetic algorithms simulate the process of natural selection which means those species who can adapt to changes in their environment are able to survive and reproduce and go to next generation. In simple words, they simulate “survival of the fittest” among individual of consecutive generation for solving a problem. Each generation consist of a population of individuals and each individual represents a point in search space and possible solution

Genetic Algorithms are being widely used in different real-world applications, for example, Designing electronic circuits, code-breaking, image processing, and artificial creativity.

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What is a Genetic Algorithm?

let's first understand basic terminologies to better understand this algorithm:

- **Population:** Population is the subset of all possible or probable solutions, which can solve the given problem.
- **Chromosomes:** A chromosome is one of the solutions in the population for the given problem, and the collection of gene generate a chromosome.
- **Gene:** A chromosome is divided into a different gene, or it is an element of the chromosome.
- **Allele:** Allele is the value provided to the gene within a particular chromosome.
- **Fitness Function:** The fitness function is used to determine the individual's fitness level in the population. It means the ability of an individual to compete with other individuals. In every iteration, individuals are evaluated based on their fitness function.
- **Genetic Operators:** In a genetic algorithm, the best individual mate to regenerate offspring better than parents. Here genetic operators play a role in changing the genetic composition of the next generation.
- **Selection**

How Genetic Algorithm Work?

The genetic algorithm works on the evolutionary generational cycle to generate high-quality solutions. These algorithms use different operations that either enhance or replace the population to give an improved fit solution.

It basically involves five phases to solve the complex optimization problems, which are given as below:

- **Initialization**
- **Fitness Assignment**
- **Selection**
- **Reproduction**
- **Termination**

1. Initialization

The process of a genetic algorithm starts by generating the set of individuals, which is called population. Here each individual is the solution for the given problem. An individual contains or is characterized by a set of parameters called Genes. Genes are combined into a string and generate chromosomes, which is the solution to the problem. One of the most popular techniques for initialization is the use of random binary strings.

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Fitness Assignment

Fitness function is used to determine how fit an individual is? It means the ability of an individual to compete with other individuals. In every iteration, individuals are evaluated based on their fitness function. The fitness function provides a fitness score to each individual. This score further determines the probability of being selected for reproduction. The high the fitness score, the more chances of getting selected for reproduction.

3. Selection

The selection phase involves the selection of individuals for the reproduction of offspring. All the selected individuals are then arranged in a pair of two to increase reproduction. Then these individuals transfer their genes to the next generation.

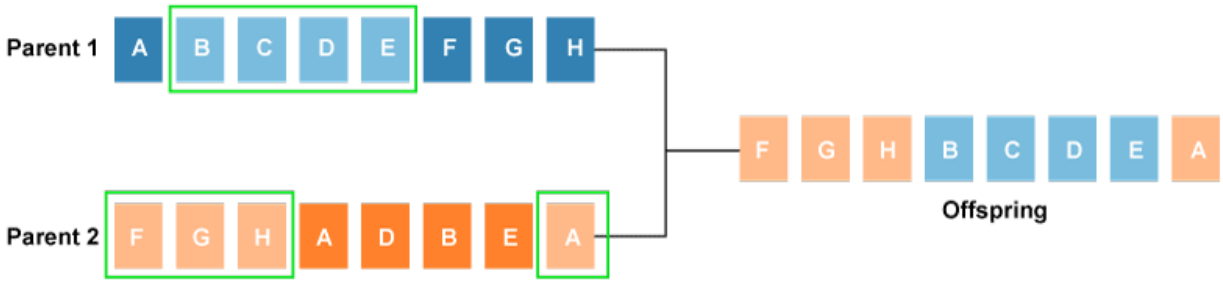
There are three types of Selection methods available, which are:

- Roulette wheel selection
- Tournament selection
- Rank-based selection

4. Reproduction

After the selection process, the creation of a child occurs in the reproduction step. In this step, the genetic algorithm uses two variation operators that are applied to the parent population. The two operators involved in the reproduction phase are given below:

- **Crossover:** The crossover plays a most significant role in the reproduction phase of the genetic algorithm. In this process, a crossover point is selected at random within the genes. Then the crossover operator swaps genetic information of two parents from the current generation to produce a new individual representing the offspring.



The genes of parents are exchanged among themselves until the crossover point is met. These newly generated offspring are added to the population. This process is also called or crossover. Types of crossover styles available:

- One point crossover
- Two-point crossover
- Livery crossover

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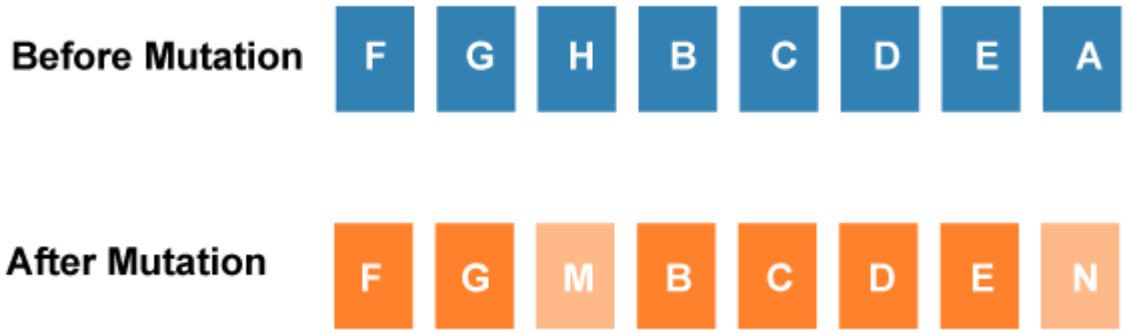
- Inheritable Algorithms crossover

Mutation

The mutation operator inserts random genes in the offspring (new child) to maintain the diversity in the population. It can be done by flipping some bits in the chromosomes. Mutation helps in solving the issue of premature convergence and enhances diversification. The below image shows the mutation process:

Types of mutation styles available,

- **Flip bit mutation**
- **Gaussian mutation**
- **Exchange/Swap mutation**



5. Termination

After the reproduction phase, a stopping criterion is applied as a base for termination. The algorithm terminates after the threshold fitness solution is reached. It will identify the final solution as the best solution in the population.

Advantages of Genetic Algorithm

- The parallel capabilities of genetic algorithms are best.
- It helps in optimizing various problems such as discrete functions, multi-objective problems, and continuous functions.
- It provides a solution for a problem that improves over time.
- A genetic algorithm does not need derivative information.

Limitations of Genetic Algorithms

- Genetic algorithms are not efficient algorithms for solving simple problems.
- It does not guarantee the quality of the final solution to a problem.
- Repetitive calculation of fitness values may generate some computational challenges.

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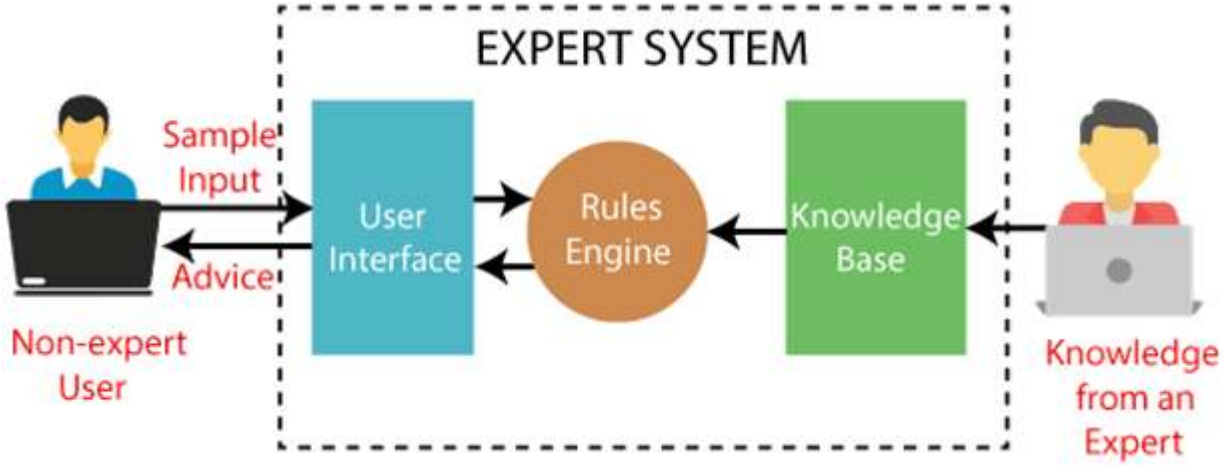
Expert Systems:

An expert system is a computer program that is designed to solve complex problems and to provide decision-making ability like a human expert. It performs this by extracting knowledge from its knowledge base using the reasoning and inference rules according to the user queries.

It solves the most complex issue as an expert by extracting the knowledge stored in its knowledge base. The system helps in decision making for compsex problems using **both facts and heuristics like a human expert**. It is called so because it contains the expert knowledge of a specific domain and can solve any complex problem of that particular domain. These systems are designed for a specific domain, such as **medicine, science,**



Characteristics of Expert System

- **High Performance:** The expert system provides high performance for solving any type of complex problem of a specific domain with high efficiency and accuracy.
- **Understandable:** It responds in a way that can be easily understandable by the user. It can take input in human language and provides the output in the same way.
- **Reliable:** It is much reliable for generating an efficient and accurate output.
- **Highly responsive:** ES provides the result for any complex query within a very short period of time.

Components of Expert System

An expert system mainly consists of three components:

- **User Interface**
- **Inference Engine**
- **Knowledge Base**

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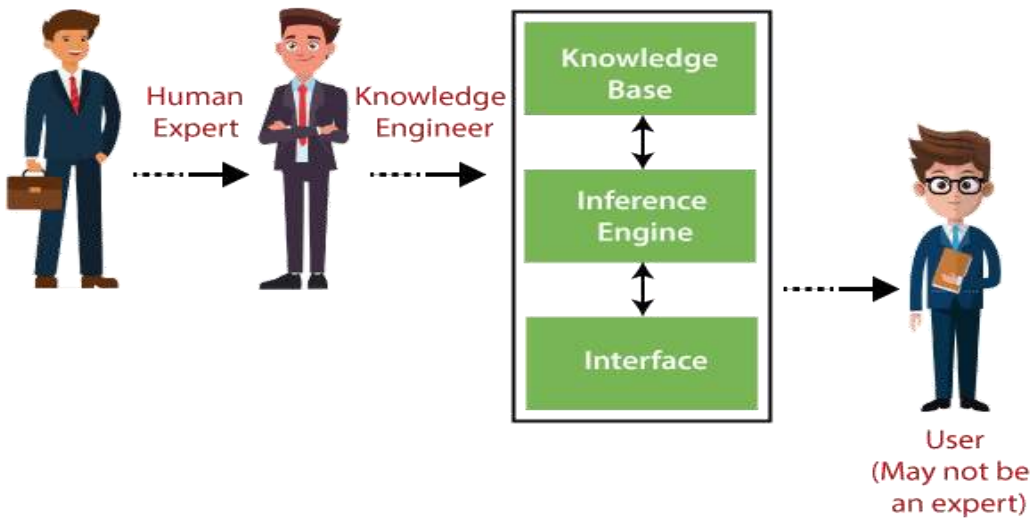

1. User Interface

With the help of a user interface, the expert system interacts with the user, takes queries as an input in a readable format, and passes it to the inference engine. After getting the response from the inference engine, it displays the output to the user. In other words, **it is an interface that helps a non-expert user to communicate with the expert system to find a solution**.

2. Inference Engine(Rules of Engine)

- The inference engine is known as the brain of the expert system as it is the main processing unit of the system. It applies inference rules to the knowledge base to derive a conclusion or deduce new information. It helps in deriving an error-free solution of queries asked by the user.
- With the help of an inference engine, the system extracts the knowledge from the knowledge base.
- There are two types of inference engine:
- **Deterministic Inference engine:** The conclusions drawn from this type of inference engine are assumed to be true. It is based on **facts** and **rules**.
- **Probabilistic Inference engine:** This type of inference engine contains uncertainty in conclusions, and based on the probability.

Inference engine uses the below modes to derive the solutions:

- **Forward Chaining:** It starts from the known facts and rules, and applies the inference rules to add their conclusion to the known facts.
- **Backward Chaining:** It is a backward reasoning method that starts from the goal and works backward to prove the known facts.

3. Knowledge Base

- The knowledgebase is a type of storage that stores knowledge acquired from the different experts of the particular domain. It is considered as big storage of knowledge. The more the knowledge base, the more precise will be the Expert System.
- It is similar to a database that contains information and rules of a particular domain or subject.

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- One can also view the knowledge base as collections of objects and their attributes. Such as a Lion is an object and its attributes are it is a mammal, it is not a domestic animal, etc.

Components of Knowledge Base

- **Factual Knowledge:** The knowledge which is based on facts and accepted by knowledge engineers comes under factual knowledge.
- **Heuristic Knowledge:** This knowledge is based on practice, the ability to guess, evaluation, and experiences.

Knowledge Representation: It is used to formalize the knowledge stored in the knowledge base using the If-else rules.

Knowledge Acquisitions: It is the process of extracting, organizing, and structuring the domain knowledge, specifying the rules to acquire the knowledge from various experts, and store that knowledge into the knowledge base.

Knowledge representation

It is the method used to organize and formalize the knowledge in the knowledge base. It is in the form of IF-THEN-ELSE rules.

Knowledge Acquisition

The success of any expert system majorly depends on the quality, completeness, and accuracy of the information stored in the knowledge base.

The knowledge base is formed by readings from various experts, scholars, and the **Knowledge Engineers**. The knowledge engineer is a person with the qualities of empathy, quick learning, and case analyzing skills.

He acquires information from subject expert by recording, interviewing, and observing him at work, etc. He then categorizes and organizes the information in a meaningful way, in the form of IF-THEN-ELSE rules, to be used by interference machine. The knowledge engineer also monitors the development of the ES.

Advantages of Expert System

- These systems are highly reproducible.
- They can be used for risky places where the human presence is not safe.
- Error possibilities are less if the KB contains correct knowledge.
- The performance of these systems remains steady as it is not affected by emotions, tension, or fatigue.
- They provide a very high speed to respond to a particular query.

Limitations of Expert System

- The response of the expert system may get wrong if the knowledge base contains the wrong information.

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- Like a human being, it cannot produce a creative output for different scenarios.
- Its maintenance and development costs are very high.
- Knowledge acquisition for designing is much difficult.
- For each domain, we require a specific ES, which is one of the big limitations.
- It cannot learn from itself and hence requires manual updates.

Applications of Expert System

- **In designing and manufacturing domain**
  It can be broadly used for designing and manufacturing physical devices such as camera lenses and automobiles.
- **In the knowledge domain**
  These systems are primarily used for publishing the relevant knowledge to the users. The two popular ES used for this domain is an advisor and a tax advisor.
- **In the finance domain**
  In the finance industries, it is used to detect any type of possible fraud, suspicious activity, and advise bankers that if they should provide loans for business or not.
- **In the diagnosis and troubleshooting of devices**
  In medical diagnosis, the ES system is used, and it was the first area where these systems were used.
- **Planning and Scheduling**
  The expert systems can also be used for planning and scheduling some particular tasks for achieving the goal of that task.

Expert System Shells

Expert system shells are toolkits that can be used to develop expert systems. They consist of some built expert system components with an empty knowledge base. Hence, in most cases, the knowledge engineer is left with only populating the knowledge base. It is essentially a special-purpose tool that is built-in in line with the requirements and standards of a particular domain or expert-knowledge area applications. It may be defined as a software package that facilitates the building of knowledge-based expert systems by providing a knowledge representation scheme and an inference engine.

Expert System Shell Structure

The Expert System Shell refers to a software module containing an:

1. User interface (built-in)
2. Inference engine (built-in)
3. A structured skeleton of a knowledge base (in its empty state) with the suitable knowledge representation facilities

Not only the above components but also some ES Shells provide facilities for database connectivity through interpreter, web integration, and natural language processing (NLP) features.

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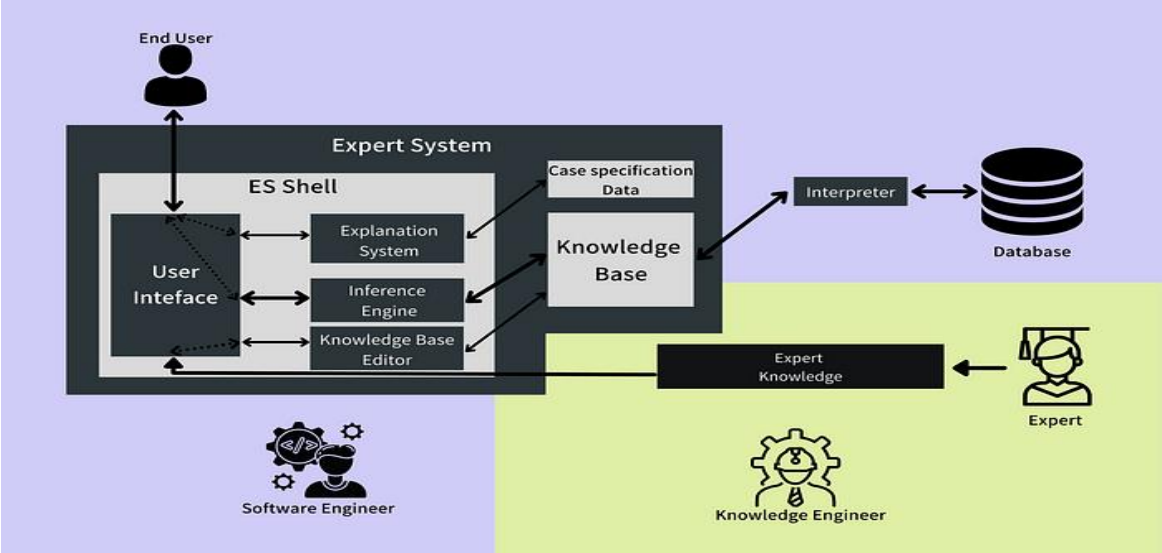

The user interface is the portal available for both end-users (who use the expert system to get solutions) and the knowledge engineer (who perform the knowledge engineering and modelling).

The knowledge base can be connected with an external database (like MySQL) since the knowledge base is not optimal in storing extensive data. The knowledge base cannot directly access the database and these access features are mediated through an interpreter.

Some ES Shells have inbuilt knowledge base editors which facilitate the Knowledge Engineer to easily update and check the knowledge base. Knowledge Engineer collects the expertise knowledge in a specific domain and models in populating the knowledge base.

Inference engine which is the most important part of an expert system access the knowledge base and solves the problem by either backward chaining or forward chaining of facts and rules in the knowledge base. In ES Shells, the inference engine is also a built-in component that is usually programmed in Prolog.

Most ES shells are composed of another component called 'Explanation System' which provides the user with reasons and explanations to provide a certain answer, by considering the 'case specification data' available.

In an expert system shell, the design of the user interface and other software components are programmed by the software engineer. Therefore, an expert system is a collaborative design of 03 major parties: expert, knowledge engineer and software engineer. (Depending on the size these parties may vary from individuals to large teams)

What is the main advantage of expert system shells?

The advantage of expert system shells was that they were somewhat easier for nonprogrammers to use. The advantage of Prolog environments was that they were not focused only on if-then rules; Prolog environments provided a much better realization of a complete first-order logic environment.

(Subject In-charge)

Prof.S.B.Mehta)

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